

An Accountable GAKA Protocol With Changeable Thresholds and Verifiable Shares in UAVs-assisted IoVs for Emergency Rescue

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Abstract—Unmanned aerial vehicles (UAVs) equipped with high line-of-sight communications have been explored as a complement to emergency communication vehicles (ECVs), particularly when ground Internet of Vehicles (IoVs) is dysfunctional resulting from natural disasters. To protect the communication security and integrity of this air-ground integrated networks created in an untrusted and open wireless environment, authentication and key agreement (AKA) is an essential mechanism to establish a secure communication channel by negotiating a session key. Nonetheless, general AKA protocols are typically not ideal for time-sensitive and computing-intensive UAVs-assisted IoVs emergency rescue, in that they are unable to simultaneously satisfy efficiency and key security requirements including accountability, resilience, thresholds changeability, shares verifiability, group adaptability, and key update. To address this challenge, this paper proposes an accountable group authentication and key agreement (GAKA) protocol with changeable thresholds and verifiable shares supporting adaptive group memberships and updatable keys (i.e. SecER). To safeguard accurate rescue decisions for ground ECVs enabled by reliable collaboration among multiple UAVs, we propose a pseudonym mechanism that aims to provide accountability for UAVs. To achieve resilience and thresholds changeability, our SecER utilizes secret sharing coupled with random parameters to seamlessly transform our GAKA protocol into one running with a new threshold in one round (i.e. round-optimal), rather than two rounds. Finally, verifiable parameters and updatable keys are respectively applied to counter deception attacks caused by maliciously distributed shares and to support adaptive network topology (i.e. UAVs joining and leaving). Extensive simulations show that compared

to the state-of-the-art approaches, our SecER is superior in balancing security and efficiency.

Index Terms—Internet of Vehicles, secure communication, unmanned aerial vehicles, group, authentication, key agreement.

I. INTRODUCTION

INTERNET of Vehicles (IoVs) integrated with computation, communication, and sensing is recognized as the most potential paradigm in the intelligent transportation system. However, poor performance of IoVs is incurred by natural disasters [1]–[3], such as wildfires, earthquakes, and tornadoes. Specifically, massive amounts of information (e.g., traffic conditions, road conditions, and rescue decisions) are disrupted due to the dysfunction of ground communication infrastructures caused by natural disasters [4].

Recently, unmanned aerial vehicles (UAVs) become a complementary method to ground emergency communication vehicles (ECVs), partly due to their ease in deploying, high flexibility, and high line-of-sight. For example, UAV groups (or swarms) can act as relay nodes of air-based networks and collaboratively assist ECVs with emergency rescue and disaster recovery. The ECV can then utilize data collected by UAVs to make optimal decisions and scheduling for emergency rescue operations. One challenge, however, is to ensure the communication security and integrity of such air-ground integrated networks considering the untrusted and open wireless operating environment. For example, how do we ensure confidentiality (e.g., highly exposed air-ground links make it easy for attackers to intercept and eavesdrop on messages) and integrity? Also, how do we mitigate risks of attackers exploiting vulnerabilities and remotely controlling UAVs to launch attacks against ground ECVs. These issues reinforce the importance of designing robust and efficient solutions to support secure communications in UAVs-assisted IoVs for emergency rescue.

There have been attempts to design authentication and key agreement (AKA) protocols to address these challenges. However, traditional point-to-point AKA protocols [5]–[11] are not designed for emergency rescue settings since these protocols focus on secure communication between a single UAV and another single UAV or between a single UAV and an ECV. One should note that a single UAV can only perform simple tasks and has limited communication coverage due to

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its limited hardware and software conditions, and this is not reflective of UAV group-empowered IoVs emergency rescue in the real-world environment. Furthermore, conventional point-to-point AKA protocols [5]–[11] generally incur excessive communication rounds and overheads. For instance, once the ECV wants to establish secure communication with n UAVs, it must repeat the point-to-point AKA protocol n times, giving rise to communication rounds and overheads required by the point-to-point AKA protocol for n times.

One could then look to group authentication and key agreement (GAKA) protocols. When designing a GAKA protocol, our core focus is on group session key security requirements and efficiency. Firstly, a threshold secret sharing can be leveraged to ensure that the GAKA protocol remains available (or resilient) even if several UAVs are offline or disconnected from the network. Equally important is that a changeable threshold can support strong security guarantees against increased attack capabilities of attackers as time goes on. Note that changeable thresholds are also necessary for UAV groups with diverse tasks (e.g. communication, computation, and sensing), because the difficulty of differentiated tasks can vary leading to different thresholds to recover secret (i.e. group session key). Thirdly, verifiable shares play an essential role in resisting deception attacks launched by the trustworthy ground ECVs (i.e. group manager), which refers to incorrect secret shares being distributed to air UAVs (i.e. group members) causing secret recovery failure. Fourthly, accountability enables identity privacy protection of legitimate UAVs and traceability of malicious UAVs. Fifthly, an adaptive emergency rescue network permitting UAVs to frequently join or leave will strengthen the practicability of our proposed GAKA protocol. Next, updatable keys are more suitable for the adaptive air-ground integrated emergency rescue network and enhance the security level of keys. Last but not least, our designed GAKA protocol must be overhead-lightweight communication, cost-effective storage, and overhead-friendly computation on the premise of security. The system framework and challenges are shown in Fig. 1.

Nevertheless, state-of-the-art GAKA protocols [12]–[28] can't simultaneously realize resilience, thresholds changeability, shares verifiability, accountability, group adaptability, key update, and high efficiency. Specifically, related works [12]–[19] require all group members must synchronize (i.e. online and connected), which means that the GAKA protocol will be unavailable for UAV-assisted IoVs emergency rescue, as long as any one of the UAVs in the group is offline incurred by entering sleep mode to extend battery life or disconnected caused by unstable wireless links and unpredictable environments. The above issue (i.e. low resilience) is solved by threshold secret sharing [20], that is to say, the secret is converted into n shares and n shares are distributed to n participants, then t (i.e. threshold) participants collaborate to restore the secret, where $t < n$.

Not surprisingly, GAKA protocols coupled with threshold secret sharing [21]–[28] cannot be directly utilized in UAVs group-empowered IoVs for emergency rescue. Except for the reference [28], the thresholds in other references are immutable, leading to security vulnerabilities when attackers'

attack capabilities improve. The neglect of shares validation in studies [21]–[28] naturally exacerbate deception attacks. Further to say, solutions [26] and [28] don't achieve accountability. Key update can't be supported in the research [26], which is only suitable for emergency rescue with the static group (i.e. group members involved in key agreement are determined in the initialization) and the static key (i.e. the same group session key in each round). The scheme [27] faces excessive overheads, even though it realizes key update.

As far as we know, no existing work simultaneously settles all the aforementioned open issues in a single protocol. In this paper, we propose an accountable GAKA protocol with changeable thresholds and verifiable shares supporting adaptive group members and updatable keys (i.e. SecER). The main contributions of our SecER are as follows:

- 1) The existing secret sharing must determine the threshold in the initialization phase (i.e. the first round). The threshold modification is achieved by discarding the previously stored initial shares, and then reinitializing the threshold (i.e. the second round). Radically different from the existing research, our SecER ameliorates the fixed threshold in secret sharing to a more flexible threshold. In other words, instead of two rounds, our SecER utilizes random parameters that blur the initial shares to seamlessly transform the GAKA protocol into one running with a new threshold in one round.
- 2) Independent of the trustworthy secret distributor (i.e. group manager), verifiable parameters are introduced to counter deception attacks by verifying the validity of shares distributed by the group manager. Further, accountability is achieved through a pseudonym mechanism, which protects the identity of legitimate UAVs while exposes the real identity of malicious UAVs. Finally, to enhance the feasibility and security, our SecER respectively implements adaptive group (i.e. UAVs joining and leaving) and updatable group session key (i.e. forward security and backward security).
- 3) Extensive simulations are executed. Firstly, under the widely used Real-Or-Random (ROR) [29] model, we prove that SecER satisfies semantic security based on elliptic curve computational Diffie-Hellman (ECCDH) and elliptic curve discrete logarithm problem (ECDLP). Secondly, compared with state-of-the-art solutions, our SecER resists common attacks and meets 12 design goals. At last, we evaluate the feasibility of SecER in terms of communication overheads, storage cost, and computational overheads.

The remainder of our paper is organized as follows. Section II reviews existing AKA protocols. Primitives are elaborated in Section III. Next, Section IV introduces models and design goals. Our proposed SecER is shown in Section V. Sections VI and VII respectively present security and performance evaluation. Finally, Section VIII concludes our research.

II. RELATED WORKS

In this section, existing works are reviewed, including point-to-point AKA protocols, GAKA protocols, and GAKA

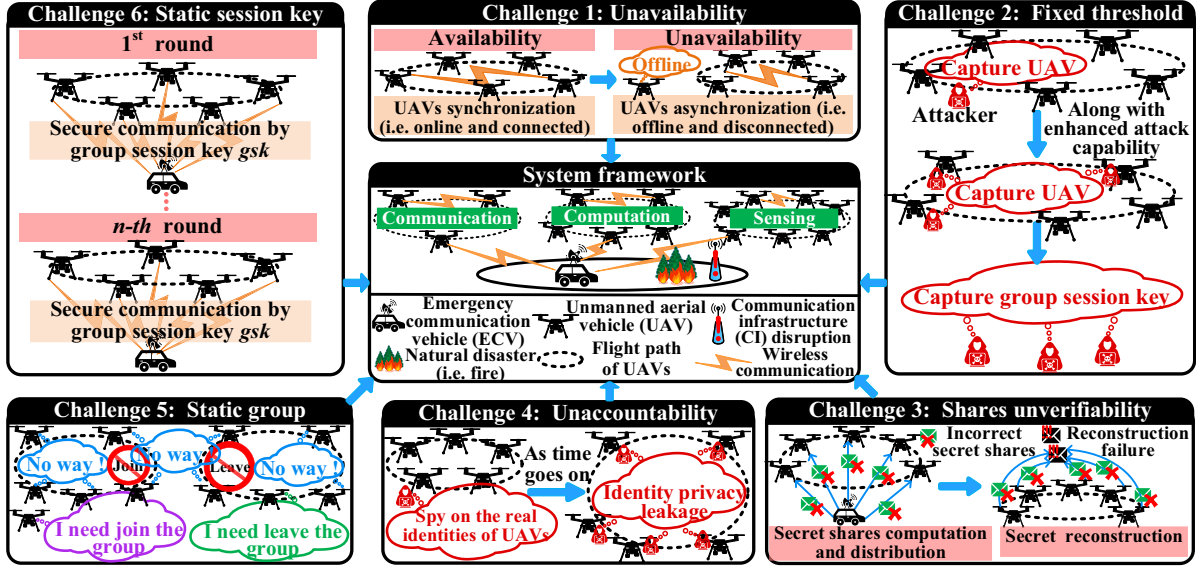


Fig. 1. System framework and challenges

protocols-based secret sharing. Afterward, as shown in Table I, we summarize the strengths and limitations of related works.

A. Point-to-point AKA Protocols

To safeguard vehicles to leave the disaster area after roadside units are interrupted by natural disasters, Cui et al. [5] proposed a two-factor AKA protocol integrated with smart cards and passwords for UAV-enabled vehicular ad hoc networks (VANETs), whereas the security level of the scheme is degraded when smart cards are lost. Considering that the availability and security are respectively limited by battery consumption and identity leakage, a lightweight privacy-preserving AKA protocol based on hash functions and XOR operation [6] was utilized for secure communication between UAVs and a zone server and between UAVs and UAVs. Note that the scheme consumes seven rounds to complete key agreement and is vulnerable to replay attacks. Combined with biometrics, passwords, and smart cards, Jiang et al. [7] designed a three-factor AKA protocol for cloud-based VANETs, to cope with common attacks and remote control by attackers.

Despite biometric encryption [8] being introduced to protect biometrics privacy, the solution is still unable to resist offline dictionary attacks and consumes large communication overheads. Such challenges were addressed by employing new primitives known as fuzzy-verifier and honeywords to a three-factor AKA protocol [9], which supports biometrics update, passwords update, and smart cards revocation. Because of the physical secure properties of physical unclonable function (PUF) (i.e. any minor disturbance to PUF resulting in inconsistent output and authentication failure), Cui et al. [10] proposed a three-factor AKA protocol with PUF and chaotic map to defend against physical attacks on autonomous vehicles. Inspired by single-point failure caused by a single trusted authority (TA) and high latency incurred by cloud-based VANETs, Wei et al. [11] proposed a privacy-preserving AKA protocol coupled with elliptic curve cryptosystem (ECC)

and pseudo-random function for fog-assisted VANETs with multi-TA. Nonetheless, the research enhances the security level at the expense of storage cost, because authentication depends on the list of valid identities stored locally by each sub-TA.

In short, the aforementioned studies [5]–[11] focus on point-to-point AKA, where each UAV executes authentication and session key agreement with each ECV separately. It's totally impractical to apply [5]–[11] directly to emergency rescue because a single UAV can only perform simple tasks and has limited communication coverage, which is not reflective of UAV group-empowered IoVs emergency rescue in the real-world environment. Additionally, communication rounds and overheads can't be underestimated.

B. GAKA Protocols

The fifth-generation cellular technology boom followed, Shang et al. [12] proposed an AKA protocol based on certificateless cryptography (CLC) and ECC for group-oriented device-to-device communication with large communication overheads. In response to multiple vehicles enjoying multiple services simultaneously, Zhang et al. [13] proposed a many-to-many AKA protocol, which claims that session key security can be guaranteed even if temporary information is leaked. However, a security vulnerability was found by Jiang et al. [14], where the update of intermediate parameters was ignored during each session, leading attackers to successfully negotiate the session key with the cloud service provider by impersonating as a legitimate vehicle. Independent of local communication infrastructure, a GAKA protocol enabled by CLC and Chinese remainder theorem [15] to achieve secure, flexible, and three-dimensional network coverage, which utilizes tethered UAVs as a substitute for dysfunctional local fog servicers. The drawback is that bilinear pairing is frequently utilized, leading to excessive computational overheads.

Encouraged by the fact that communication overheads of a short signature are less than half of that traditional signature,

TABLE I
STATE-OF-THE-ART WORKS COMPARISON

Category	Scheme	Year	Primitives	Advantages	Limitations
Point-to-point AKA protocols	Cui et al. [5]	2023	Smart cards, passwords, and chaotic map.	Prevent password-guessing attacks and realize password update.	The security level is degraded when smart cards are lost.
	Pu et al. [6]	2022	Physically unclonable function (PUF), hash function, and XOR operation.	Resist various well-known attacks and have low computation costs.	Consume seven rounds to complete the key agreement and be vulnerable to replay attacks.
	Jiang et al. [7]	2020	Biometric extraction, hash function, and XOR operation.	Cope with remote control by attackers.	Communication overheads are large.
	Yang et al. [8]	2016	Biometric encryption and bilinear pairing.	Protect biometrics privacy.	Be vulnerable to offline dictionary attacks.
	Wang et al. [9]	2023	Hash function, fuzzy extractor, and XOR operation.	Support password update, biometrics update, and smart card revocation.	Computational overheads are large.
	Cui et al. [10]	2023	PUF and chaotic map.	Defend against physical attacks on autonomous vehicles.	Session keys are easy to leakage due to missing key update.
	Wei et al. [11]	2023	Elliptic curve cryptosystem (ECC).	Deal with single-point failure caused by a single trusted authority (TA).	Deal with single-point failure caused by a single trusted authority (TA).
GAKA protocols	Shang et al. [12]	2020	ECC and certificateless cryptography (CLC).	Achieve secure device-to-device group communication.	Communication overheads are large.
	Zhang et al. [13]	2020	ECC, hash function, and XOR operation.	Realize secure many-to-many communication.	The update of intermediate parameters is ignored during each session.
	Jiang et al. [14]	2022	ECC, hash function, and XOR operation.	Resist impersonation attacks.	More communication rounds are consumed.
	Tan et al. [15]	2023	CLC and Chinese remainder theorem.	Independent of local communication infrastructure, achieve secure, flexible, and three-dimensional network coverage.	Bilinear pairing is frequently utilized, leading to excessive computational overheads.
	Yang et al. [16]	2023	Bilinear pairing, ECC, and Boneh-Lynn-Shacham signature.	Design a short signature-enabled GAKA protocol to allow memberships to join or leave.	The scheme is low performance including frequent broadcasting messages and multiple rounds of sessions.
	Wu et al. [17]	2018	Zero-knowledge proof, paillier encryption, and identity-based cryptography (IBC).	Realize ownership authentication between a user and mobile devices.	It's only suitable for secure communication between two mobile devices and a user.
	Guo et al. [18]	2022	Hash function and XOR operation.	Achieve secure communication between multiple (i.e. greater than or equal to three) mobile devices and a user.	Forward security is compromised because group proof and session key aren't updated after wearable devices leaving or revocation.
	Xie et al. [19]	2024	Hash function, XOR operation, and symmetric cryptography.	Undermine the foundation of TA.	Certificate management issues remain unresolved.
GAKA protocols based on secret sharing	Tong et al. [24]	2021	Low-density parity-check code, secret sharing, and IBC.	Distributed key management with efficient rounds of interactions for space networks.	Computational overheads of key update are large.
	Zhang et al. [25]	2021	Fuzzy extractor, PUF, and secret sharing.	Ensure the privacy of biometrics stored in multiple servers while achieving identity authentication.	Attackers can ultimately obtain the secret even if they aren't involved in secret recovery.
	Yang et al. [26]	2022	Bilinear pairing, hash function, threshold secret sharing, and XOR operation.	Realize the access authentication and cross-domain authentication.	The binding between the registration status and vehicle identity brings about identity leakage.
	Nakkar et al. [27]	2023	Threshold secret sharing, hash function, and message authentication code.	Enable batch authentication and resist asynchronous release attacks.	The group has to be static.
	Zhang et al. [28]	2023	Threshold secret sharing, chaotic map, and hash function.	Assist the disconnected UAV obtain the latest group key.	The security of key is threatened because of the static UAV identity.
Our scheme	SecER	2024	Hash function, threshold secret sharing, bilinear pairing, and ECC.	Satisfy 12 design goals related to security and have more preponderances in balancing security and availability.	Storage costs are slightly higher.

Yang et al. [16] constructed a BLS short signature-enabled GAKA protocol to allow memberships to join or leave, whereas this scheme is low performance (i.e. frequent broadcasting messages and multiple rounds of sessions). Aiming at the ownership of two mobile devices from a user, Wu et al. [17] proposed a secure two-party computation AKA protocol integrating zero-knowledge proof and paillier encryption into an identity-based cryptosystem (IBC), which isn't suitable for secure communication between multiple (i.e. greater than or equal to three) mobile devices and a user. Such an issue is addressed by Guo et al. [18], but forward security is compromised because group proof and session key aren't updated after wearable devices leaving or revocation. To undermine the foundation of TA, Xie et al. [19] divided servers into registration servers, authentication servers, and computing servers. Despite this study allowing semi-trusted authentication servers to assist key agreement, certificate management issues remain unresolved.

In summary, the aforementioned studies [12]–[19] emphasize that with the assistance of the group manager, group members negotiate a group session key in an open wireless network to ensure secure communication between the group manager and group members. Perhaps unsurprisingly, GAKA protocols [12]–[19] aren't a sensible strategy for time-sensitive and computing-intensive emergency rescue, as they require all group members must synchronize (i.e. online or connected). In other words, solutions [12]–[19] will be unavailable for UAV-assisted IoVs emergency rescue, if any one of the UAVs in the group is asynchronous (i.e. offline or disconnected).

C. GAKA Protocols-Based Secret Sharing

Shamir first proposed secret sharing in 1979 [20]. As a burgeoning solution for secure distributed data storage, secret sharing is broadly supportive of researchers. In 2013, Harn first introduced secret sharing into group authentication [21].

Drawing lessons from this idea, secret sharing was subsequently applied to ring signature [22] and IBC [23] to realize group-oriented authentication and key agreement.

Tong et al. [24] introduced secret sharing into low-density parity-check code so that cluster heads efficiently calculate keys. However, computational overheads of key update is large when the satellite joins a new domain.

Zhang et al. [25] designed cancelable biometrics based on fuzzy extractor and physical unclonable function. This scheme ensures the privacy of biometrics stored in multiple servers while achieving identity authentication, whereas it has a security flaw that the attacker can ultimately obtain the secret even if he isn't involved in secret recovery.

Yang et al. [26] proposed the access authentication and cross-domain authentication protocols for edge computing-based VANETs. Nonetheless, the binding between the registration status and vehicle identity brings about identity leakage. Besides, the frequent use of bilinear mapping leads to high computational overheads.

Nakkar et al. [27] introduced threshold secret sharing for edge nodes, which enable batch authentication and resist asynchronous release attacks. However, the scheme requires the group to be static, leading to the unavailability of the protocol when group members would like to join or leave.

Zhang et al. [28] proposed a threshold-based GAKA protocol, which utilizes variable polynomial coefficients and random generators to alter key contributions. The security level is markedly affected by the randomness of seeds required in random generators. Furthermore, key security is threatened when a disconnected UAV can still calculate the updated group key because of the static UAV identity.

In conclusion, state-of-the-art GAKA protocols based on secret sharing [20]–[28] cannot be directly utilized for UAVs-assisted emergency rescue. Specifically, except for the reference [28], the threshold of others is immutable, which leads to a sharp drop in the security level of the protocol as the adversary's attack capability increases and isn't flexibly applied to air-ground integrated emergency rescue. The neglect of shares validation in related works [20]–[28] naturally exacerbate deception attacks. Further to say, studies [24], [26], [28] don't achieve accountability and key update. Solutions [26] and [27] face excessive overheads, even though they realize key update.

III. PRELIMINARIES

In this section, we elaborate on preliminaries used in SecER, including bilinear pairing, hard problems, and (t, n) -Shamir secret sharing.

A. Bilinear Pairing

Assume that G_1 is additive cyclic groups with the prime order q on elliptic curve (ECC) and G_2 is multiplicative cyclic groups. Bilinear pairing [30] $e : G_1 \times G_1 \rightarrow G_2$ satisfies the following properties.

1) **Bilinearity**: For any $R, \mathbb{R}, Q \in G_1$ and $\lambda, \zeta \in \mathbb{Z}_q^*$, there are

$$\begin{cases} e(\lambda R, \zeta Q) = e(\zeta R, \lambda Q) = e(R, Q)^{\lambda \zeta} = e(Q, R)^{\lambda \zeta} \\ e(R + \mathbb{R}, Q) = e(R, Q)e(\mathbb{R}, Q) \end{cases}$$

2) **Computability**: For any $R, Q \in G_1$, bilinear pairing $e(R, Q)$ is effectively computed.

3) **Non-degeneracy**: $e(R, Q) \neq 1$ for all $R, Q \in G_1$.

B. Hard Problems

u, v, w, z are random numbers selected from $\mathbb{Z}_q^*$ and the elliptic curve is denoted by $E_{u,v} : y^2 = x^3 + ux + v$, where u and v satisfy $4u^3 + 27v^2 \neq 0 \pmod{p}$. P is the generator of additive cyclic group G_1 .

1) **Elliptic curve computational Diffie-Hellman (ECCD-H) problem**: Known wP and zP , then wzP can be calculated.

2) **Elliptic curve discrete logarithm problem (ECDLP)**: Known $P, Q = wP$, w can be calculated, where $Q \in G_1$.

C. (t, n) -Shamir Secret Sharing

Let THS and SES respectively represent threshold space and secret space, where $THS, SES \in \mathbb{Z}_q^*$. n is the total number of members and t is the threshold for secret recovery. (t, n) -Shamir secret sharing [20] includes the following algorithms.

1) **Shares generation**: Suppose the polynomial of degree $t - 1$ is $f(x) = d_0 + d_1x + d_2x^2 + \dots + d_{t-1}x^{t-1} \pmod{p}$, where the polynomial coefficient of $f(x)$ is $d_k \in \mathbb{Z}_q^* (k = 0, 1, 2, \dots, t - 1)$ and the shared secret is $se = f(0) = d_0$. $share_i = f(x_i)$ denotes the share of each member, where $i = 1, 2, \dots, n$. Note that the above parameters are defined from a macro perspective in this section and they will be instantiated in the construction of SecER (i.e. Section V). To make the paper more rigorous and avoid confusion among readers, we use different symbols to distinguish the definition in this section from the instantiation in Section V.

2) **Secret recovery**: Based on the Lagrange interpolation polynomial $l_i(x) = \prod_{j=1, j \neq i}^t \frac{x - x_j}{x_i - x_j}$ and secret shares of t members (i.e. $(x_1, share_1), (x_2, share_2), \dots, (x_t, share_t)$), the polynomial of degree $t - 1$ can be denoted as $f(x) = d_0 + d_1x + d_2x^2 + \dots + d_{t-1}x^{t-1} \pmod{p} = \sum_{i=1}^t share_i l_i(x) = \sum_{i=1}^t share_i \prod_{j=1, j \neq i}^t \frac{x - x_j}{x_i - x_j}$. Therefore, the shared secret se is restored by calculating $se = f(0) = \sum_{i=1}^t share_i \prod_{j=1, j \neq i}^t \frac{-x_j}{x_i - x_j}$.

IV. NETWORK MODELS AND DESIGN GOALS

In this section, we introduce network models involving the system model, security model, and threat model, in which the system model expounds on network entities. Next, the security model and threat model are respectively security proof fundamental and cyberspace threats definition. There follow by design goals.

A. System Model

The system model contains the ECV and UAVs, which are defined in detail as follows.

1) **ECV**: The ECV utilizes data collected by UAVs to make optimal decisions and scheduling for emergency rescue.

2) **UAVs**: As a substitute for terrestrial communication infrastructures damaged by natural disasters, UAVs are subdivided into three groups and different groups of UAVs play different roles. Specifically, communication UAVs provide downlink communication for rescuers. Sensing UAVs collect data from mobile and wearable devices of survivors. Computing UAVs predict locations of survivors.

B. Security Model

The extensively used Real-Or-Random (ROR) model [29] has been utilized to the security proof of AKA protocols [11], [13], [18], [25], [28]. In SecER, session key security is proved by the ROR model, which is defined as follows.

Participants: Let $\mathbb{I}$ denote the instance. $\mathbb{I}_{ECV}^i$ and $\mathbb{I}_{UAV}^j$ are respectively denote the i -th instance of the ECV and the j -th instance of the UAV.

Queries: The following queries implement the interaction between the adversary and the participant.

- $Execute(\mathbb{I}_{ECV}^i, \mathbb{I}_{UAV}^j)$: Messages between the ECV and UAVs are passively eavesdropped by the adversary $\mathcal{A}$ through $Execute(\mathbb{I}_{ECV}^i, \mathbb{I}_{UAV}^j)$.
- $Send(\mathbb{I}, mes)$: To simulate active attacks, $\mathcal{A}$ gets the response after sending a forged message mes .
- $Reveal(\mathbb{I})$: $\mathcal{A}$ gets the current session key through the query.
- $Test(\mathbb{I}_{ECV}^i)$: Session key security (i.e. semantic security) is proved by this query, which is executed once but can be executed at any time. The output of $Test(\mathbb{I}_{ECV}^i)$ is represented as a random bit $\lambda \in \{0, 1\}$. $\mathcal{A}$ will receive a random string as long as the session key only if $\lambda = 0$, or else $\mathcal{A}$ will receive the session key. $\perp$ is sent to $\mathcal{A}$ if $\mathbb{I}_{ECV}^i$ doesn't output a session key.

Accepted state: Once the instance $\mathbb{I}$ receives expected messages, $\mathbb{I}$ goes into the accepted state.

Freshness: If $\mathcal{A}$ cannot obtain the session key by the $Reveal()$ [31], then $\mathbb{I}_{ECV}^i$ and $\mathbb{I}_{UAV}^j$ are fresh. $Reveal()$ [31] refers to communication channel being captured by $\mathcal{A}$, who can eavesdrop, forge, delete, and inject messages.

In the ROR model, $\mathcal{A}$ wins the game only if the guessed bit λ' by $\mathcal{A}$ is equal to the outputted bit λ by $Test()$.

C. Threat Model

SecER utilizes the Dolev-Yao (DY) threat model [32], which permits attackers to eavesdrop, tamper, forge, inject, and delete messages, meaning that attackers can thoroughly control the public communication channel. Given that Canetti and Krawczyk's adversary model (CK-adversary model) [33] is regarded as the standard model for AKA protocols, our SecER also utilizes the CK-adversary model, which can break forward security (i.e. permitting attackers to get the previous session keys). Note that the ECV and UAVs are semi-trusted, that is to say, the ECV may distribute incorrect secret shares to UAVs to launch deception attacks. Furthermore, UAVs are curious about the identities of legal UAVs.

D. Design Goals

According to the system model, security model, and threat model, we propose design goals DG_i .

- 1) **DG_1 Anonymity**: Attackers can neither speculate the real identity of UAVs, nor two sessions sent by the same UAV.
- 2) **DG_2 Unforgeability**: Adversaries cannot forge messages that can successfully pass registration and authentication.
- 3) **DG_3 Mutual authentication**: To ensure communication security, the ECV and UAVs must have mutual authentication before establishing a communication connection.
- 4) **DG_4 Semantic security**: A shared group session key is negotiated between the ECV and UAVs.
- 5) **DG_5 Key update**: Due to dynamic network topology caused by UAVs joining or leaving the group, the group session key must be updated.
- 6) **DG_6 Resilience**: Our proposed scheme remains available (or resilient) even if several UAVs are offline or disconnected from the network.
- 7) **DG_7 Accountability**: The real identity of legitimate UAVs is protected, but the real identity of UAVs is disclosed if UAVs send forged perception data and the wrong location of survivors or UAVs are remotely controlled by attackers.
- 8) **DG_8 Thresholds changeability**: It's essential to have changeable thresholds, since different UAV groups need different thresholds for secret (i.e. group session key) recovery. In addition, changeable thresholds can resist adversaries with stronger attack capabilities, that is, by increasing the threshold of secret recovery to counter the increasing number of captured group members (i.e. UAVs) in the group.
- 9) **DG_9 Shares verifiability**: To prevent deception attacks (i.e. the ECV sending incorrect shares, resulting in failed secret recovery), UAVs need to verify secret shares distributed by the ECV.
- 10) **DG_{10} Forward security**: Newly joined UAVs can't infer the previous session key based on the current session key.
- 11) **DG_{11} Backward security**: The leaving UAVs can't infer the future session key based on the current session key.
- 12) **DG_{12} Efficiency**: Overheads (i.e. computational overheads and communication overheads) and storage cost need to be minimized on the basis of ensuring security.

V. PROPOSED SECER

In this section, we recap our proposed SecER and expound the construction of SecER.

A. Outline

Our SecER involves setup, UAVs register, GAKA, shares computation, thresholds changeability, shares verifiability, key recovery, and group adaptivity. The details are as follows and the interaction process of SecER shows in Fig. 2.

- 1) **Setup**: The ECV generates long-term keys and ephemeral keys.
- 2) **UAVs register**: UAVs first register their identities, then pseudonyms are generated with the assistance of the ECV.
- 3) **GAKA**: After the ECV and UAVs complete mutual authentication, the group session key is generated for secure

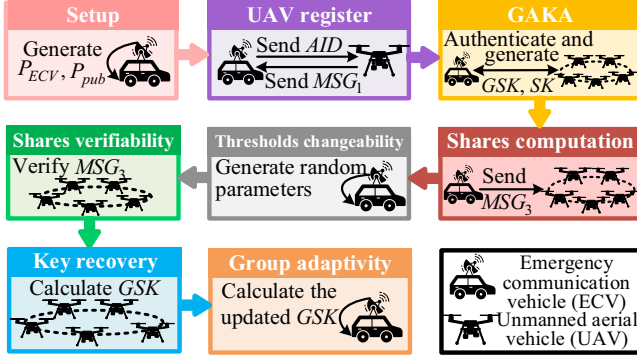


Fig. 2. The interaction process of SecER

communication among UAVs within groups (or swarms). Subsequently, the session key is negotiated to ensure secure communication between a single UAV and the ECV.

4) **Shares computation**: The ECV calculates secret shares, which are then distributed to UAVs.

5) **Thresholds changeability**: Random parameters are generated to change thresholds.

6) **Shares verifiability**: Each UAV verifies the distributed share by the verifiable parameter.

7) **Key recovery**: UAVs collaborate to calculate the group session key.

8) **Group adaptivity**: The group session key is updated by the ECV for UAVs joining or leaving.

B. Our proposed Scheme SecER

Key symbols are presented in Table II and the implementation process of SecER shows in Fig. 3. In fact, the calculation processes for the UAV computation group, UAV communication group, and UAV sensing group are similar during setup, UAVs register, GAKA, shares computation, thresholds changeability, shares verifiability, key recovery, and group adaptivity. The slight difference is that Mem_j^{comp} is replaced with Mem_j^{comm} or Mem_j^{sense} for setup, UAVs register, GAKA, shares computation, shares verifiability, and group adaptivity. Therefore, we take the UAV computation group as an example to elaborate in detail during setup, UAVs register, GAKA, shares computation, shares verifiability, and group adaptivity.

1) Setup

Given the additive cyclic group G_1 on elliptic curve with the order q and multiplicative cyclic group G_2 . The generators of G_1 are P and P_1 . Bilinear pairing is defined as $e : G_1 \times G_1 \rightarrow G_2$. Moreover, one-way hash functions are defined as $H_1 : G_1 \rightarrow \{0, 1\}^*$, $H_2 : G_1 \times \{0, 1\}^* \rightarrow \{0, 1\}^*$, and $H_3 : G_1 \times \{0, 1\}^* \times \{0, 1\}^* \rightarrow Z_q^*$. MapToPoint hash functions are defined as $H_4 : G_1 \times \{0, 1\}^* \times \{0, 1\}^* \times G_1 \times \{0, 1\}^* \rightarrow G_1$ and $H_5 : \{0, 1\}^* \times G_1 \times \{0, 1\}^* \rightarrow G_1$. The ECV randomly selects b_i and s_i from Z_q^* as ephemeral private key and long-term private key, respectively. At last, the ECV calculates ephemeral public keys $P_{ECV} = b_i P$ and long-term public key $P_{pub} = s_i P$.

2) UAVs Register

TABLE II
KEY SYMBOLS

Symbol	Description
ECV	Emergency communication vehicle ECV
$UAV_j^{comp}, UAV_j^{comm}, UAV_j^{sense}$	Computing UAV UAV_j^{comp} , communication UAV UAV_j^{comm} , and sensing UAV UAV_j^{sense}
$Mem_j^{comp}, Mem_j^{comm}, Mem_j^{sense}$	The member of computing UAV, communication UAV, and sensing UAV
$AID_{Mem_j^{comp}}, AID_{Mem_j^{comm}}, AID_{Mem_j^{sense}}$	The pseudonyms of $Mem_j^{comp}, Mem_j^{comm}$, and Mem_j^{sense}
$G_1, G_2; e$	Additive cyclic group, multiplicative cyclic group, bilinear pairing
$a_{Mem_j^{comp}}, b_i$	Ephemeral private keys of Mem_j^{comp} and ECV
$P_{Mem_j^{comp}}, P_{ECV}$	Ephemeral public keys of Mem_j^{comp} and ECV
$s_{Mem_j^{comp}}, s_i$	Long-term private keys of Mem_j^{comp} and ECV
$S_{Mem_j^{comp}}, P_{pub}$	Long-term public keys of Mem_j^{comp} and ECV
SK_{ij}/SK_{ji}	Session key between ECV and Mem_j^{comp}
GSK	Group session key
$\oplus; ; H_i(i) = 1, 2, 3, 4, 5$	XOR operation, concatenation, hash functions
$\Delta T_j; \Delta T_{ECV}$	The validity period of $AID_{Mem_j^{comp}}$ and ID_{ECV}
$share$	The secret share
t, n, m	The initial threshold, total number of group members, updated threshold
$V_1, V_2, \dots, V_{t-1}$	Verifiable parameters
$RP_1, RP_2, \dots, RP_m$	Random parameters

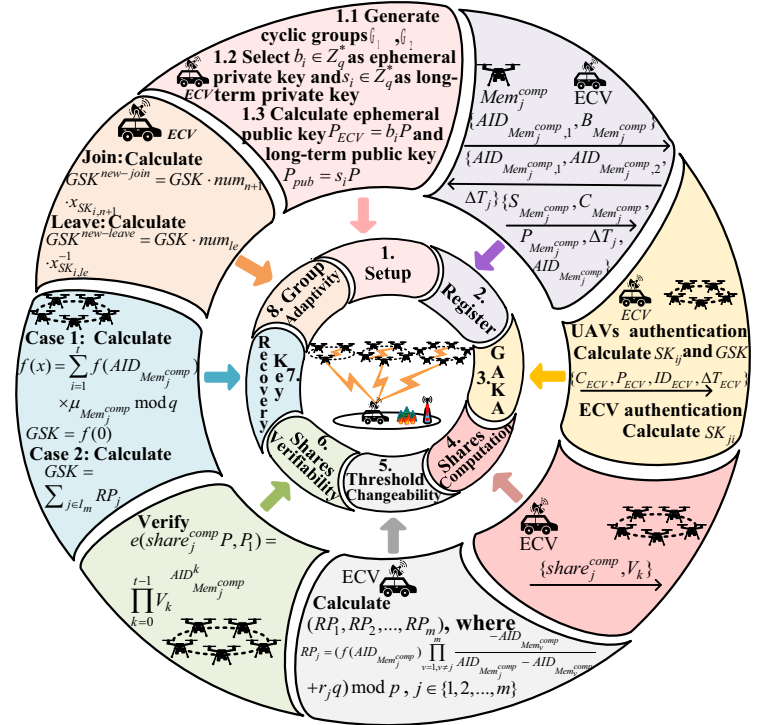


Fig. 3. The implementation process diagram of SecER

Step 1: The member of the UAV computation group (denoted by Mem_j^{comp}) randomly selects $\alpha_{Mem_j^{comp}} \in Z_q^*$ and the real identity $RID_{Mem_j^{comp}}$ to calculate $AID_{Mem_j^{comp},1} = \alpha_{Mem_j^{comp}}P$ and $B_{Mem_j^{comp}} = H_1(\alpha_{Mem_j^{comp}}P_{pub}) \oplus RID_{Mem_j^{comp}}$, then $\{AID_{Mem_j^{comp},1}, B_{Mem_j^{comp}}\}$ are sent to the ECV.

Step 2: After receiving $\{AID_{Mem_j^{comp},1}, B_{Mem_j^{comp}}\}$, the ECV calculates $RID_{Mem_j^{comp}} = B_{Mem_j^{comp}} \oplus H_1(s_i AID_{Mem_j^{comp},1})$ and $AID_{Mem_j^{comp},2} = RID_{Mem_j^{comp}} \oplus H_2(s_i AID_{Mem_j^{comp},1} || \Delta T_j)$. In the end, the ECV stores $AID_{Mem_j^{comp}} = \{AID_{Mem_j^{comp},1}, AID_{Mem_j^{comp},2}, \Delta T_j\}$ locally and then the ECV sends $AID_{Mem_j^{comp}}$ to Mem_j^{comp} .

Step 3: Mem_j^{comp} randomly selects $a_{Mem_j^{comp}}$ and $s_{Mem_j^{comp}}$ from Z_q^* as ephemeral private key and long-term private key, respectively. Subsequently, Mem_j^{comp} calculates ephemeral public keys $P_{Mem_j^{comp}} = a_{Mem_j^{comp}}P$ and long-term public key $S_{Mem_j^{comp}} = s_{Mem_j^{comp}}P$. Finally, Mem_j^{comp} sends $MSG_1 = \{S_{Mem_j^{comp}}, C_{Mem_j^{comp}}, P_{Mem_j^{comp}}, AID_{Mem_j^{comp}}, \Delta T_j\}$ to the ECV, where $C_{Mem_j^{comp}} = s_{Mem_j^{comp}}H_4(AID_{Mem_j^{comp}} || P_{Mem_j^{comp}} || \Delta T_j)$.

3) GAKA

Step 1: After the ECV receives $\{\{S_{Mem_1^{comp}}, C_{Mem_1^{comp}}, P_{Mem_1^{comp}}, AID_{Mem_1^{comp}}, \Delta T_1\}, \{S_{Mem_2^{comp}}, C_{Mem_2^{comp}}, P_{Mem_2^{comp}}, AID_{Mem_2^{comp}}, \Delta T_2\}, \dots, \{S_{Mem_n^{comp}}, C_{Mem_n^{comp}}, P_{Mem_n^{comp}}, AID_{Mem_n^{comp}}, \Delta T_n\}\}$, UAVs' identities validity is checked by the equation $e(\sum_{j=1}^n C_{Mem_j^{comp}}, P) = \prod_{j=1}^n e(H_4(AID_{Mem_j^{comp}} || P_{Mem_j^{comp}} || \Delta T_j), S_{Mem_j^{comp}})$. Step 2 is performed if the above equation holds, otherwise GAKA is termination. Correctness proof is as follows.

$$\begin{aligned} & e(\sum_{j=1}^n C_{Mem_j^{comp}}, P) \\ &= e(s_{Mem_1^{comp}}H_4(AID_{Mem_1^{comp}} || P_{Mem_1^{comp}} || \Delta T_1) + s_{Mem_2^{comp}}H_4(AID_{Mem_2^{comp}} || P_{Mem_2^{comp}} || \Delta T_2) + \dots + s_{Mem_n^{comp}}H_4(AID_{Mem_n^{comp}} || P_{Mem_n^{comp}} || \Delta T_n), P) \\ &= e(s_{Mem_1^{comp}}H_4(AID_{Mem_1^{comp}} || P_{Mem_1^{comp}} || \Delta T_1), P) \cdot e(s_{Mem_2^{comp}}H_4(AID_{Mem_2^{comp}} || P_{Mem_2^{comp}} || \Delta T_2), P) \cdot \dots \cdot e(s_{Mem_n^{comp}}H_4(AID_{Mem_n^{comp}} || P_{Mem_n^{comp}} || \Delta T_n), P) \\ &= e(H_4(AID_{Mem_1^{comp}} || P_{Mem_1^{comp}} || \Delta T_1), S_{Mem_1^{comp}}) \cdot e(H_4(AID_{Mem_2^{comp}} || P_{Mem_2^{comp}} || \Delta T_2), S_{Mem_2^{comp}}) \cdot \dots \cdot e(H_4(AID_{Mem_n^{comp}} || P_{Mem_n^{comp}} || \Delta T_n), S_{Mem_n^{comp}}) \\ &= \prod_{j=1}^n e(H_4(AID_{Mem_j^{comp}} || P_{Mem_j^{comp}} || \Delta T_j), S_{Mem_j^{comp}}) \end{aligned} \quad (1)$$

Step 2: The ECV calculates session key with $Mem_j^{comp}(j \in \{1, 2, \dots, n\})$ denoted as $SK_{ij} = b_i P_{Mem_j^{comp}} = (x_{SK_{ij}}, y_{SK_{ij}})$. Note that $x_{SK_{ij}}$ and $y_{SK_{ij}}$ are respectively x-coordinate and y-coordinate of SK_{ij} . Subsequently, the ECV calculates group session key represented as $GSK = \prod_{j=1}^n x_{SK_{ij}}P = XP$, where

$$X = \prod_{j=1}^n x_{SK_{ij}}.$$

Step 3: The ECV calculates $C_{ECV} = s_i H_5(ID_{ECV} || P_{ECV} || \Delta T_{ECV})$. Soon afterwards, $MSG_2 = \{C_{ECV}, P_{ECV}, ID_{ECV}, \Delta T_{ECV}\}$ is sent to Mem_j^{comp} .

Step 4: After receiving $MSG_2 = \{C_{ECV}, P_{ECV}, ID_{ECV}, \Delta T_{ECV}\}$, the Mem_j^{comp} authenticates the identity validity of ECV by the equation $e(C_{ECV}, P) = e(H_5(ID_{ECV} || P_{ECV} || \Delta T_{ECV}), P_{pub})$. Next, Mem_j^{comp} calculates the session key between Mem_j^{comp} and ECV (indicated as $SK_{ji} = a_{Mem_j^{comp}}P_{ECV} = (x_{SK_{ji}}, y_{SK_{ji}})$) if the above equation holds, otherwise GAKA is termination. Note that $x_{SK_{ji}}$ and $y_{SK_{ji}}$ are respectively x-coordinate and y-coordinate of SK_{ji} . Correctness proof is as follows.

$$\begin{aligned} & e(C_{ECV}, P) \\ &= e(s_i H_5(ID_{ECV} || P_{ECV} || \Delta T_{ECV}), P) \\ &= e(H_5(ID_{ECV} || P_{ECV} || \Delta T_{ECV}), s_i P) \\ &= e(H_5(ID_{ECV} || P_{ECV} || \Delta T_{ECV}), P_{pub}) \end{aligned} \quad (2)$$

4) Shares Computation

Before calculating secret shares, UAVs must complete mutual authentication and session keys agreement. Assume that the polynomial denotes $g(x) = D_0 + D_1x + D_2x^2 + \dots + D_{t-1}x^{t-1} \bmod \hat{p}$, where $D_0 = g(0) = X = \prod_{j=1}^n x_{SK_{ij}}$ belongs to the finite field $Z_{\hat{q}}$ and other parameters D_k ($k = 1, 2, 3, \dots, t-1$) belongs to the finite field $Z_{\hat{p}}$. Note that prime numbers $\hat{p}$ and $\hat{q}$ satisfy $\hat{p} > \hat{q} + n\hat{q}^2$, in which n denotes the total number of group members. The ECV calculates the verifiable parameter $V_k = e(D_kP, P_1)$ ($k = 0, 1, 2, \dots, t-1$) and secret shares $share_j^{comp} = g(x_j) = g(H_3(AID_{Mem_j^{comp}}))$, where $AID_{Mem_j^{comp}}$ is obtained during mutual authentication between UAVs and the ECV. Finally, $MSG_3 = \{share_j^{comp}, V_k\}$ is sent to Mem_j^{comp} .

5) Thresholds Changeability

Suppose the initial threshold and the updated threshold for UAV computation group are respectively t and m , whereas the updated threshold for UAV communication group and UAV sensing group are respectively m^* and l , where $t \leq m, m^*, l \leq n$. Actually, thresholds changeability for the UAV computation group, UAV communication group, and the UAV sensing group is similar, just change the threshold m to m^* and l to calculate the corresponding parameters. For ease of understanding, we take thresholds changeability for the UAV computation group as an example and specific details are as follow.

The ECV selects m shares of the UAV communication group expressed as $(share_1^{comp}, share_2^{comp}, \dots, share_m^{comp})$ and random numbers $r_j \in Z_{\hat{q}}$ to calculate corresponding random parameters $(RP_1, RP_2, \dots, RP_m)$, where $RP_j = (g(H_3(AID_{Mem_j^{comp}})) \prod_{v=1, v \neq j}^m \frac{-H_3(AID_{Mem_v^{comp}}) - H_3(AID_{Mem_j^{comp}})}{H_3(AID_{Mem_j^{comp}}) - H_3(AID_{Mem_v^{comp}})} + r_j\hat{q}) \bmod \hat{p}$, $j \in \{1, 2, \dots, m\}$. Note that prime numbers $\hat{p}$ and $\hat{q}$ satisfy $\hat{p} > \hat{q} + n\hat{q}^2$, in which n denotes the total number of UAV computation group members.

6) Shares Verifiability

After receiving $MSG_3 = \{share_j^{comp}, V_k\}$, shares are verified by the equation $e(share_j^{comp}P, P_1) = \prod_{k=0}^{t-1} H_3^k(AID_{Mem_j^{comp}}) \cdot V_k$. Shares distributed by the ECV are valid if the above equation holds, or else the protocol is terminated. Correctness proof is as follows.

$$\begin{aligned}
e(share_j^{comp}P, P_1) &= e(g(H_3(AID_{Mem_j^{comp}}))P, P_1) \\
&= e((D_0 + D_1H_3(AID_{Mem_j^{comp}}) + D_2H_3^2(AID_{Mem_j^{comp}}) + \dots + D_{t-1}H_3^{t-1}(AID_{Mem_j^{comp}}) \bmod \hat{p})P, P_1) \\
&= e((D_0 + D_1H_3(AID_{Mem_j^{comp}}) + D_2H_3^2(AID_{Mem_j^{comp}}) + \dots + D_{t-1}H_3^{t-1}(AID_{Mem_j^{comp}}))P, P_1) \\
&= e(D_0P, P_1)e(D_1P, P_1) \dots e(D_{t-1}P, P_1) \\
&= V_0^{H_3^0(AID_{Mem_j^{comp}})} V_1^{H_3^1(AID_{Mem_j^{comp}})} V_2^{H_3^2(AID_{Mem_j^{comp}})} \dots V_{t-1}^{H_3^{t-1}(AID_{Mem_j^{comp}})} \\
&= \prod_{k=0}^{t-1} V_k^{H_3^k(AID_{Mem_j^{comp}})}
\end{aligned} \tag{3}$$

Note that prime number $\hat{p}$ is bigger than both the total number of group members (denoted by n) and $g(H_3(AID_{Mem_j^{comp}}))$.

7) Key Recovery

Case 1 The threshold is t : t UAVs collaboratively compute

$$g(x) = \sum_{j=1}^t g(H_3(AID_{Mem_j^{comp}})) \mu_{Mem_j^{comp}} \bmod \hat{p}, \text{ where}$$

$$\mu_{Mem_j^{comp}} = \prod_{k=1, k \neq j}^t \frac{x - H_3(AID_{Mem_k^{comp}})}{H_3(AID_{Mem_j^{comp}}) - H_3(AID_{Mem_k^{comp}})}.$$

The group session key is $GSK = g(0) \cdot P$.

Case 2 The threshold is m : m UAVs collaboratively compute the group session key $GSK = (\sum_{j=1}^m RP_j \bmod \hat{q}) \cdot P$, where $t \leq m \leq n$. Correctness proof is as follows.

$$\begin{aligned}
&\sum_{j=1}^m RP_j \bmod \hat{q} \\
&= \sum_{j=1}^m (g(H_3(AID_{Mem_j^{comp}})) \prod_{v=1, v \neq j}^m \frac{-H_3(AID_{Mem_v^{comp}})}{H_3(AID_{Mem_j^{comp}}) - H_3(AID_{Mem_v^{comp}})} + r_j \hat{q}) \bmod \hat{p} \bmod \hat{q} \\
&= (g(0) + \sum_{j=1}^m r_j \hat{q}) \bmod \hat{p} \bmod \hat{q}
\end{aligned} \tag{4}$$

Due to $g(0), r_j \in Z_{\hat{q}}$, we can know $\sum_{j=1}^m r_j \hat{q} \leq \sum_{j=1}^n r_j \hat{q} < n\hat{q} = n\hat{q}^2$, which can deduce $(g(0) + \sum_{j=1}^m r_j \hat{q}) < \hat{q} + n\hat{q}^2$. Considering that prime numbers $\hat{p}$ and $\hat{q}$ satisfy $\hat{p} > \hat{q} + n\hat{q}^2$, we further know $(g(0) + \sum_{j=1}^m r_j \hat{q}) < \hat{q} + n\hat{q}^2 < \hat{p}$ and deduce $\sum_{j=1}^m RP_j \bmod \hat{q} = (g(0) + \sum_{j=1}^m r_j \hat{q}) \bmod \hat{p} \bmod \hat{q} = (g(0) + \sum_{j=1}^m r_j \hat{q}) \bmod \hat{q} = g(0) = X$. Therefore, the group session key is $GSK = g(0) \cdot P$.

8) Group Adaptivity

Case 1 UAVs Joining: Assume that newly joined UAV is denoted as Mem_{n+1}^{comp} , then registration phase is performed to obtain ephemeral private key $a_{Mem_{n+1}^{comp}}$ and ephemeral public key $P_{Mem_{n+1}^{comp}} = a_{Mem_{n+1}^{comp}}P$. Afterwards, the ECV selects $num_{n+1} \in Z_q^*$ to calculate the updated group session key $GSK^{new-join} = GSK \cdot num_{n+1} \cdot x_{SK_{i,n+1}}$, the updated session key

$SK_{i,n+1} = b_i P_{Mem_{n+1}^{comp}} = (x_{SK_{i,n+1}}, y_{SK_{i,n+1}})$. The ECV sends $\{C_{ECV}^{new}, P_{ECV}^{new}, ID_{ECV}, \Delta T_{ECV}^{new}\}$ to Mem_{n+1}^{comp} , which calculates the updated session key $SK_{n+1,i} = a_{Mem_{n+1}^{comp}} P_{ECV} = (x_{SK_{n+1,i}}, y_{SK_{n+1,i}})$.

Case 2 UAVs Leaving: UAVs leaving is similar to UAVs joining, we will not elaborate on it here. The implementation process is briefly described as follows. Assuming that newly leaved UAV is represented as Mem_{le}^{comp} , the ECV selects $num_{le} \in Z_q^*$ to calculate the updated session key $GSK^{new-leave} = GSK \cdot num_{le} \cdot x_{SK_{i,le}}^{-1}$.

VI. SECURITY EVALUATION

In this section, we evaluate the security of our SecER in terms of security proof and security analysis. Based on the ROR model, we prove that our SecER is semantic security (i.e. session key security) under ECCDH and ECDLP. Moreover, to emphasize strengths, we conduct a security analysis between state-of-the-art protocols and our SecER.

A. Security Proof

Theorem 1 Let $\mathcal{A}$ indicate the adversary who breaches semantic security (i.e. session key security) in polynomial time pt . Then, the advantage that $\mathcal{A}$ breaches semantic security in SecER is indicated as follows

$$\begin{aligned}
Adv_{\mathcal{A}}^{SecER}(pt) &\leq \frac{q_{H_3}^2}{|H_3|} + \frac{q_{H_4}^2}{|H_4|} + \frac{q_{H_5}^2}{|H_5|} + Adv_{\mathcal{A}}^{ECCDH}(pt) \\
&\quad + Adv_{\mathcal{A}}^{ECDLP}(pt)
\end{aligned}$$

where q_{H_i} and $|H_i|$ are respectively the times of hash queries and the space of hash function, where $i = 3, 4, 5$. $Adv_{\mathcal{A}}^{ECCDH}(pt)$ and $Adv_{\mathcal{A}}^{ECDLP}(pt)$ are respectively the advantage of solving ECCDH problem and the advantage of solving ECDLP in polynomial time pt .

Proof. $\Pr[Suc_{GM_i}]$ and Adv_{GM_i} respectively denote the probability and advantage of $\mathcal{A}$ winning games GM_i ($i = 1, 2, 3, 4, 5$), where $\Pr[Suc_{GM_i}] = Adv_{GM_i}$.

Game GM_1 : Based on the random oracle model, GM_1 simulates that SecER is breached. Therefore,

$$Adv_{\mathcal{A}}^{SecER}(pt) = |2 \Pr[Suc_{GM_1}] - 1| \tag{5}$$

Game GM_2 : $\mathcal{A}$ eavesdrops $MSG_1 - MSG_3$ by $Execute(\mathbb{I}_{ECV}^j, \mathbb{I}_{UAV}^j)$ queries in GM_2 . Furthermore, GM_2 has the same advantages as GM_1 . That is,

$$\Pr[Suc_{GM_2}] = \Pr[Suc_{GM_1}] \tag{6}$$

Game GM_3 : Active attacks are launched by $Send()$ queries. Given that the probability of hash collisions is almost negligible, $MSG_1 - MSG_3$ are associated with random numbers, pseudonyms, and timestamps. Based on birthday paradox, we get

$$|\Pr[Suc_{GM_3}] - \Pr[Suc_{GM_2}]| \leq \frac{q_{H_3}^2}{|H_3|} + \frac{q_{H_4}^2}{|H_4|} + \frac{q_{H_5}^2}{|H_5|} \tag{7}$$

Game GM_4 : In GM_4 , known $P_{Mem_j^{comp}} = a_{Mem_j^{comp}}P$ and $P_{ECV} = b_i P$ (where $a_{Mem_j^{comp}}, b_i \in Z_q^*$), the probability that $\mathcal{A}$ calculates session key (i.e. $SK_{ji} = a_{Mem_j^{comp}} b_i P$) is

$$|\Pr[Suc_{GM_4}] - \Pr[Suc_{GM_3}]| \leq Adv_{\mathcal{A}}^{ECCDH}(pt) \tag{8}$$

Game GM_5 : In GM_5 , the prerequisite for $\mathcal{A}$ to capture the group session key (i.e. $GSK = \prod_{j=1}^n x_{SK_{ij}}P$) is to obtain $X = \prod_{j=1}^n x_{SK_{ij}}$. Thus, the probability of $\mathcal{A}$ capturing the group session key is as follows.

$$|\Pr[Suc_{GM_5}] - \Pr[Suc_{GM_4}]| \leq Adv_A^{ECDLP}(pt) \quad (9)$$

After games GM_i ($i = 1, 2, 3, 4, 5$), the probability that $\mathcal{A}$ guesses the output bit correctly is $\frac{1}{2}$, so

$$\Pr[Suc_{GM_5}] = \frac{1}{2} \quad (10)$$

Based on equations (5)(6)(10), we can know

$$\begin{aligned} \frac{1}{2} Adv_A^{SecER}(pt) &= |\Pr[Suc_{GM_1}] - \frac{1}{2}| \\ &= |\Pr[Suc_{GM_2}] - \Pr[Suc_{GM_5}]| \end{aligned} \quad (11)$$

In view of equations (7)-(9), we get

$$\begin{aligned} |\Pr[Suc_{GM_2}] - \Pr[Suc_{GM_5}]| &\leq |\Pr[Suc_{GM_2}] - \Pr[Suc_{GM_3}]| \\ &\quad + |\Pr[Suc_{GM_3}] - \Pr[Suc_{GM_4}]| \\ &\quad + |\Pr[Suc_{GM_4}] - \Pr[Suc_{GM_5}]| \\ &\leq \frac{q_{H_3}^2}{|H_3|} + \frac{q_{H_4}^2}{|H_4|} + \frac{q_{H_5}^2}{|H_5|} \\ &\quad + Adv_A^{ECCDH}(pt) \\ &\quad + Adv_A^{ECDLP}(pt) \end{aligned} \quad (12)$$

To sum up, founded on equations (11) and (12), we know

$$\begin{aligned} Adv_A^{SecER}(pt) &\leq \frac{q_{H_3}^2}{|H_3|} + \frac{q_{H_4}^2}{|H_4|} + \frac{q_{H_5}^2}{|H_5|} \\ &\quad + Adv_A^{ECCDH}(pt) + Adv_A^{ECDLP}(pt) \end{aligned} \quad (13)$$

B. Security Analysis

The security of SecER is analyzed based on the 12 design goals defined in Section IV-D, among which the efficiency (i.e. DG_{12}) is detailed in the performance evaluation (i.e. Section VII). Table III compares the security of our SecER with state-of-the-art studies [11], [12], [15], [26], [28], to indicate the merits of our SecER.

TABLE III
COMPARISON OF SECURITY

Security	Ref [11]	Ref [12]	Ref [15]	Ref [26]	Ref [28]	SecER
DG_1	✓	✓	✓	✓	✓	✓
DG_2	✓	✓	✓	✓	✓	✓
DG_3	✓	✓	✓	✓	✓	✓
DG_4	✓	×	×	×	✓	✓
DG_5	×	✓	✓	✓	✓	✓
DG_6	×	×	×	✓	✓	✓
DG_7	✓	×	✓	×	×	✓
DG_8	×	×	×	×	✓	✓
DG_9	×	×	×	×	×	✓
DG_{10}	×	✓	✓	✓	✓	✓
DG_{11}	×	✓	✓	✓	✓	✓
DG_{12}	×	✓	✓	×	✓	✓

1) **DG_1 Anonymity:** Given $AID_{Mem_j^{comp},1} = \alpha_{Mem_j^{comp}}P$ and $P_{pub} = s_iP$, attackers is unable to calculate $\alpha_{Mem_j^{comp}}P_{pub}$ due to ECCDH problem, leading to real identities calculation of UAVs failed.

2) **DG_2 Unforgeability:** To forge messages, adversaries must simultaneously know UAVs' ephemeral private key and long-term private key, which is contradictory to the ECDLP.

3) **DG_3 Mutual authentication:** On the one hand, the ECV authenticates with UAVs by the equation $e(\sum_{j=1}^n C_{Mem_j^{comp}}, P) = \prod_{j=1}^n e(H_4(AID_{Mem_j^{comp}} || P_{Mem_j^{comp}} || \Delta T_j), S_{Mem_j^{comp}})$. On the other hand, the UAV authenticates the ECV by the equation $e(C_{ECV}, P) = e(H_5(ID_{ECV} || P_{ECV} || \Delta T_{ECV}), P_{pub})$.

4) **DG_4 Semantic security:** Semantic security is proved in Section VI-A.

5) **DG_5 Key update:** To improve the availability of UAV groups with dynamic network topology, SecER supports session key update for both joining and leaving UAVs.

6) **DG_6 Resilience:** Secret sharing is introduced to ensure our SecER is still available even if several UAVs are offline or disconnected from the network. Additionally, thresholds changeability supports strong security guarantees against increased attack capabilities of attackers as time goes on.

7) **DG_7 Accountability:** The ECV calculates and broadcasts the real identity of UAV (i.e. $RID_{Mem_j^{comp}} = AID_{Mem_j^{comp},2} \oplus H_2(s_i AID_{Mem_j^{comp},1} || \Delta T_j)$), as long as the UAV releases forged rescue messages.

8) **DG_8 Thresholds changeability:** In SecER, random parameters are utilized to alter the threshold, which is not only resisting the adversaries with enhanced attack capabilities (i.e. security-enhanced), but also suitable for air-ground intergrated emergency rescue (i.e. feasibility-improved).

9) **DG_9 Shares verifiability:** Deception attacks are defended by the equation $e(\text{share}_j^{comp}P, P_1) = \prod_{k=0}^{t-1} V_k^{H_3^k(AID_{Mem_j^{comp}})}$.

10) **DG_{10} Forward security:** The security of updated session key depends on confidentiality and randomness of random numbers, and ECCDH problem when a new UAV joins the group. Therefore, a newly joined UAV can only know the current session key, rather than the previous session key.

11) **DG_{11} Backward security:** Similar to forward security, based on confidentiality and randomness of random numbers, and ECCDH problem, a leaving UAV can't know the future session key according to the current session key.

12) **DG_{12} Efficiency:** Our SecER manifests some preponderances in communication overheads, storage cost, and computational overheads (see Section VII).

VII. PERFORMANCE EVALUATION

In this section, we evaluate the performance of our SecER in terms of communication overheads, storage cost, and computational overheads. For communication overheads and storage cost, it's worth noting that consumptions of the group manager (corresponding to the ECV in SecER) and group members (corresponding to UAVs in SecER) is in different phases, so we evaluate them from the perspectives of the group manager and group members. However, for computational overheads, the consumption of the group manager and group members exists at each phase, so we evaluate it based on different phases.

TABLE IV
COMMUNICATION OVERHEADS COMPARISON (IN BYTES)

Scheme	Group Member				Group Manager				
	Register	GAKA	Key update	Sum	Setup	Register	Sum	GAKA	Key update
Ref [11]	$2 Z_q^* $	$2 G_4 + 7 Hash + 4 Time + 4 Z_q^* $	—	356	0	$ CT + 2 Hash $	168	$n(2 Z_q^* + Time + 2 Hash)$	—
Ref [12]	$ ID + G_1 $	$6 ID + 3 Z_q^* + 3 Time + G_1 + 2 Hash + Sig $	$ ID + Hash $	488	0	$ Z_q^* + G_1 + ID $	158	$n(ID + Z_q^*)$	$(n+1)(ID + Z_q^*)$
Ref [15]	—	$4 Z_q^* + G_3 + Time $	$ ID + Z_q^* $	242	$4 Z_q^* $	—	80	$(4n+3) Z_q^* + (n+1) Time + n(G_3 + 3 G_2)$	$(n+1)(4 Z_q^* + G_3 + Time)$
Ref [26]	$ ID $	$2 ID + 6 G_3 + 2 Time + Z_q^* $	$2 ID + 6 G_3 + 2 Time + Z_q^* $	1642	$3 G_3 $	$ G_3 + ID $	522	$(3n+2) G_3 + (n+1) Z_q^* + n(Time + ID)$	$(n+1)(5 G_3 + 2 Z_q^* + Time + ID)$
Ref [28]	—	$ CT + Z_q^* + MAC + Time + 3 Hash $	$ CT + Z_q^* + MAC + Time + 3 Hash $	488	$3 Z_q^* $	—	60	$n(CT + Z_q^* + 2 MAC + 2 Time + 3 Hash)$	$(n+1)(CT + 3 Z_q^* + MAC + 2 Time + 3 Hash)$
SecER	$2 Time + 5 G_1 + 2 Z_q^* $	0	0	688	0	$ G_1 + Z_q^* + Time $	152	$n(2 G_1 + Time + ID)$	$(n+1)(2 G_1 + Time + ID)$

n is the number of group members.

— is there is no such phase.

A. High-Level Description

1) **Experimental environment:** To simulate cryptographic operations, based on java pairing-based cryptography (JPBC) library [34], G_1 in the bilinear pairing $e : G_1 \times G_1 \rightarrow G_2$ is the additive group over the elliptic curve $ECC_1 : y^2 = x^3 + x \mod p$, where the embedding degree is 2 and p is 512-bit prime number. Note that G_1 is generated by the generator P with 160-bit prime order q . Additionally, simulation experiments are run on the laptop (HP Windows 10, i7-10750H @ 2.6 GHz, and 16 GB memory).

2) **Comparative schemes:** To evaluate the performance systematcially, comprehensively, and objectively, we choose related works [11], [12], [15], [26], [28] as comparative schemes.

- **Ref [11]:** It is a point-to-point AKA protocol for VANETs without resilience, thresholds changeability, shares verifiability, group adaptability, and key update.
- **Ref [12]:** It is a GAKA protocol for fifth generation cellular networks without resilience, accountability, thresholds changeability, shares verifiability, and group adaptability.
- **Ref [15]:** It is a GAKA protocol for UAV-enabled infrastructure-less Internet of vehicles without resilience, thresholds changeability, and shares verifiability.
- **Ref [26]:** It is a GAKA protocol-based secret sharing for

VANETs without accountability, thresholds changeability, and shares verifiability.

- **Ref [28]:** It is a GAKA protocol-based secret sharing for UAV ad hoc networks without accountability and shares verifiability.

B. Communication Overheads

Table IV presents a comparison of communication overheads, where the size of identity ((denoted as $|ID|$)), times-tamp (denoted as $|Time|$), the element on Z_q^* (denoted as $|Z_q^*|$), the element on the addition cyclic group G_1 (denoted as $|G_1|$), the element on the multiplication cyclic group G_2 (denoted as $|G_2|$), the elements on the addition cyclic group G_3 (denoted as $|G_3|$), the element on the addition cyclic group G_4 (denoted as $|G_4|$), ciphertext (denoted as $|CT|$), message authentication code (denoted as $|MAC|$), hash function (denoted as $|Hash|$), and signature (denoted as $|Sig|$) are respectively 10bytes [12], 4bytes, 20bytes, 128bytes [15], 128bytes [26], 40bytes [11], 128bytes, 32bytes, 20bytes [28], and 20bytes [12].

A comparison of communication overheads is shown in Fig. 4. In the registration phase, the UAV sends $\{AID_{Mem_j^{comp}, 1}, B_{Mem_j^{comp}}\}$ and $\{S_{Mem_j^{comp}}, C_{Mem_j^{comp}}, P_{Mem_j^{comp}}, AID_{Mem_j^{comp}}, \Delta T_j\}$ to the ECV, where $|AID_{Mem_j^{comp}}| = |G_1| + |Z_q^*| + |Time|$,

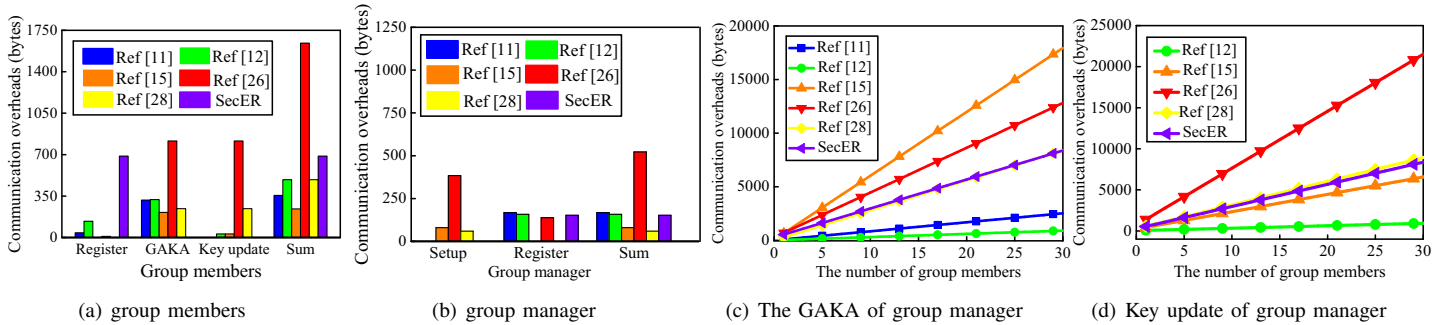


Fig. 4. Communication overheads comparison

$AID_{Mem_j^{comp},1}, S_{Mem_j^{comp}}, C_{Mem_j^{comp}}, P_{Mem_j^{comp}} \in G_1$, $B_{Mem_j^{comp}} \in Z_q^*$, and ΔT_j is the timestamp. Hence, the communication overhead of the group member (i.e. the UAV) is $5|G_1| + 2|Z_q^*| + 2|Time| = 688 \text{ bytes}$. As a response, the ECV transmits $\{AID_{Mem_j^{comp},1}, AID_{Mem_j^{comp},2}, \Delta T_j\}$ to the UAV, where $AID_{Mem_j^{comp},1} \in G_1$, $AID_{Mem_j^{comp},2} \in Z_q^*$, and ΔT_j is the timestamp. Therefore, the communication overhead of the group manager (i.e. the ECV) is $|G_1| + |Z_q^*| + |Time| = 152 \text{ bytes}$. During the setup phase, there is no messages interaction between UAVs and the ECV, so their communication overheads are zero. As for GAKA phase, the ECV transmits $\{C_{ECV}, P_{ECV}, ID_{ECV}, \Delta T_{ECV}\}$ to the UAV, where $C_{ECV}, P_{ECV} \in G_1$ and ID_{ECV} is the identity of the ECV. Consequently, the communication overhead of the ECV is $2|G_1| + |Time| + |ID| = 270 \text{ bytes}$. For key update, taking UAVs joining as an example, the ECV sends $\{C_{ECV}^{new}, P_{ECV}^{new}, ID_{ECV}, \Delta T_{ECV}^{new}\}$ to $(n+1)$ UAVs so that the communication overhead of the ECV is $(n+1)(2|G_1| + |Time| + |ID|) = 270(n+1) \text{ bytes}$. Similarly, communication overheads of references [11], [12], [15], [26], [28] can be obtained, which will not be discussed.

Fig. 4 (a) represents group members' communication overheads comparison. In SecER, UAVs send $\{AID_{Mem_j^{comp},1}, B_{Mem_j^{comp}}\}$ and MSG_1 for identity authentication in the register phase, whereas no message is sent during the GAKA phase and key update phase. As a result, group members of our SecER have very little communication overheads, which also indicates that our SecER is more suitable for resource-constrained UAVs. The root cause for excessive communication overheads in the scheme [26] is that under the same security level, communication overheads consumed by the super singular elliptic curve are greater than that of the non-singular elliptic curve.

Fig. 4 (b) shows communication overheads of the group manager in the setup phase, registration phase, and the sum. To track malicious UAVs, the ECV assists UAVs to generate pseudonyms, which is ignored in the solution [28] and is also the reason why the communication overhead of our SecER is higher than that of the solution [28]. There is no registration phase in the work [15] and the required parameters are generated directly in the setup phase, so the communication overhead of the work [15] is slightly less than our SecER. Fortunately, communication overheads of group manager in SecER is more advantageous than that of studies [11], [12], [26]. It's worth mentioning that the sum communication overheads of the group manager in our SecER has been reduced by 30% on average.

Fig. 4 (c) indicates the group manager communication overheads comparison in the GAKA phase. Studies [11], [12] have more strengths than our SecER because lightweight hash functions and XOR operation are utilized. However, our SecER has more advantages over works [15], [26], [28], which becomes more obvious with the increase of group members. For example, the communication overhead of SecER's GAKA phase is reduced by 1.744% on average for 30 UAVs.

As shown in Fig. 4 (d), we just compare communication overheads in the key update phase between our SecER and

references [12], [15], [26], [28], in that the work [11] doesn't support key update. The research [12] leverages the identity and random number to achieve key update with lightweight communication overheads. Contrarily, the study [26] utilizes expensive elements of addition cyclic group from super singular elliptic curve, leading to excessive communication overheads. Notably, our SecER has remarkable dominance along with the increase of group members. For instance, the communication overhead of SecER's key update phase is reduced by 12.44% on average for 30 UAVs.

C. Storage Cost

Table V presents a comparison of storage cost. A comparison of storage cost is represented in Fig. 5. In the setup phase, the ECV need store ephemeral private key b_i , long-term private key s_i , ephemeral public key $P_{ECV} = b_iP$, and long-term public key $P_{pub} = s_iP$, where $P_{ECV}, P_{pub} \in G_1$. Thus, the storage cost of ECV is $2(|Z_q^*| + |G_1|) = 296 \text{ bytes}$. In the register phase, UAVs and the ECV need storage $\{AID_{Mem_j^{comp},1}, AID_{Mem_j^{comp},2}, \Delta T_j\}$, where $AID_{Mem_j^{comp},1} \in G_1$, $AID_{Mem_j^{comp},2} \in Z_q^*$, and ΔT_j is the timestamp. Therefore, the storage cost of both the group manager (i.e. the ECV) and group members (i.e. UAVs) is $|G_1| + |Z_q^*| + |Time| = 152 \text{ bytes}$. Lastly, the group manager and group members also need store group session key $GSK \in G_1$ so that storage cost of the GAKA phase is $|G_1| = 128 \text{ bytes}$. In the same way, communication overheads of references [11], [12], [15], [26], [28] can be obtained.

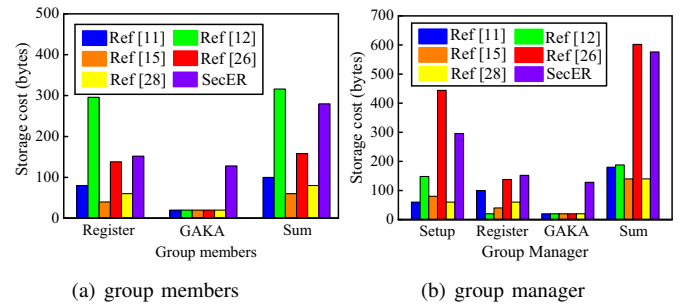


Fig. 5. Storage cost comparison

Fig. 5 (a) exhibits storage cost comparison of group members. As regards the register phase, group members store two key pairs in the work [12], leading to a higher cost than our SecER. However, references [11], [15], [26], [28] are cost-effective storage because they store more lightweight output of hash functions or elements from Z_q^* . For the GAKA phase, the session key stored in references [11], [12], [15], [26], [28] is the element from Z_q^* , while the session key stored in our SecER is the element from G_1 . Hence, our SecER requires an additional 108 bytes during the GAKA phase, which is affordable for UAVs.

Fig. 5 (b) reveals storage cost comparison of the group manager. During the setup phase, the storage cost of the group manager in our SecER is higher than that of other studies [11], [12], [15], [28], because the ECV stores more parameters including ephemeral private key, ephemeral public

TABLE V
STORAGE COST COMPARISON (IN BYTES)

Scheme	Group Member			Group Manager			
	Register	GAKA	Sum	Setup	Register	GAKA	Sum
Ref [11]	$2(Z_q^* + Hash)$	$ Z_q^* $	100	$ Z_q^* + G_4 $	$3 Z_q^* + G_4 $	$ Z_q^* $	180
Ref [12]	$2(Z_q^* + G_1)$	$ Z_q^* $	316	$ Z_q^* + G_1 $	$2 ID $	$ Z_q^* $	188
Ref [15]	$2 Z_q^* $	$ Z_q^* $	60	$4 Z_q^* $	$2 Z_q^* $	$ Z_q^* $	140
Ref [26]	$ G_3 + ID $	$ Z_q^* $	158	$3(Z_q^* + G_3)$	$ ID + G_3 $	$ Z_q^* $	602
Ref [28]	$3 Z_q^* $	$ Z_q^* $	80	$3 Z_q^* $	$3 Z_q^* $	$ Z_q^* $	140
SecER	$ G_1 + Z_q^* + Time $	$ G_1 $	280	$2(Z_q^* + G_1)$	$ G_1 + Z_q^* + Time $	$ G_1 $	576

key, long-term private key, and long-term public key to keep the freshness of session key in each communication round. In the register phase, instead of storing the identity and random numbers as in references [11], [12], [15], [26], [28], the manager of our SecER stores the pseudonym and timestamp resulting in higher storage cost. For the GAKA phase, our SecER requires an additional 108 bytes, which is affordable for the well-resourced ECV.

D. Computational Overheads

Tables VI and VII respectively show runtime of operations and comparison of computational overheads. Computational overheads at different phases are demonstrated on Fig. 6. In the setup phase, the ECV consumes $2T_{G_1-Mul}$ to calculate P_{ECV} and P_{pub} . As for the register phase, the UAV consumes $5T_{G_1-Mul} + T_{mgh} + T_h$ to calculate $AID_{Mem_j^{comp},1}$, $B_{Mem_j^{comp}}$, $P_{Mem_j^{comp}}$, $S_{Mem_j^{comp}}$, and $C_{Mem_j^{comp}}$, then consumes $T_{G_1-Mul} + 2T_h$ to calculate $RID_{Mem_j^{comp}}$ and $AID_{Mem_j^{comp},2}$. Thus, the computational overheads of register phase is $6T_{G_1-Mul} + 3T_h + T_{mph}$. In the same way, we can know the computational overheads of the GAKA phase, thresholds changeability, shares verification, secret recovery, and key update in SecER, so are schemes [11], [12], [15], [26], [28].

TABLE VI
RUNTIME OF OPERATIONS (IN MILLISECOND)

Operation	Definition	Runtime
T_{G_1-Mul}	Multiplication operation on G_1	2.6994
T_{G_2-Mul}	Multiplication operation on G_2	2.8640
T_{G_3-Mul}	Multiplication operation on G_3	2.6784
T_{G_4-Mul}	Multiplication operation on G_4	2.5394
T_{G_1-Add}	Addition operation on G_1	0.0377
T_{G_3-Add}	Addition operation on G_3	0.0342
T_{G_4-Add}	Addition operation on G_4	0.0293
$T_{Z_q^*-Mul}$	Multiplication operation on Z_q^*	0.0041
T_{bp}	Bilinear pairing	4.301
T_h	One-way hash function	0.0012
T_{mph}	MapToPoint hash function	7.5683
T_{exp}	Exponential operation	0.715
T_{mod}	Modular operation	0.626
T_{sig}	Signature	0.107
T_{prf}	Pseudo random function	0.012
T_{La}	Lagrange interpolation	0.007
T_{lookup}	Lookup operation on cuckoo filter	0.001
T_{che}	Chebyshev polynomial	0.2631
$T_{en/de}$	Encryption/Decryption	0.645

Fig. 6 (a) displays computational overheads in setup and register phases. With respect to the setup phase, time-consuming multiplication operation from G_3 is utilized in schemes [15], [26]. References [11], [12] are respectively 2.8594 milliseconds and 2.6994 milliseconds faster than our

SecER. The solution [28] is overhead-friendly computation due to generating random seeds in the setup phase, rather than time-consuming system parameters. For the register phase, the study [11] only performs hash operation on the identity and secret shares of the work [28] are calculated in advance, which are reasons for their low computational overheads at this phase. It's acceptable that the study [12] is 10.1935 milliseconds faster than our SecER. After all, our SecER supports security requirements (i.e. resilience, accountability, thresholds changeability, shares verifiability, and group adaptability) that the study [12] ignores. Additionally, computational overhead of our SecER is 13.5216 milliseconds more than that of the work [26], which ignores accountability, thresholds changeability, and shares verifiability.

As exhibited in Fig. 6 (b), the advantages of our SecER is more obvious than solutions [15], [26], whereas it's inferior to [11], [12], [28] because our SecER frequently utilizes time-consuming bilinear pairing and MapToPoint hash function to verify identity legitimacy in the GAKA phase.

As represented in Fig. 6 (c), we just compare computational overheads in the secret recovery phase between our SecER and references [26], [28], in that works [11], [12], [15] doesn't support resilience. Compared with references [26], [28], our SecER has outstanding advantages, which become more prominent with the increase of thresholds.

As shown in Fig. 6 (d), we just compare computational overheads in the key update phase between our SecER and references [12], [15], [26], [28], in that the work [11] doesn't support key update. In the key update phase, our SecER is inferior to that of solutions [12], [15], [28], while our SecER supports group adaptability ignored in the work [12], resilience and thresholds changeability overlooked in the scheme [15], accountability and shares verifiability neglected in the study [28]. Compared with the work [26], our SecER has outstanding advantages, which become more prominent with the increase of thresholds. To sum up, our SecER has more preponderances in balancing security and efficiency.

E. Discussion

Our designed SecER coupled with changeable thresholds, verifiable shares, pseudonyms, and random parameters is proposed for UAV-assisted IoVs emergency rescue. Subsequently, extensive security evaluations (including security proof and analysis) and comprehensive performance simulations (including communication overheads, storage costs, and computational overheads) have been completed, indicating that our SecER is superior in balancing security and efficiency.

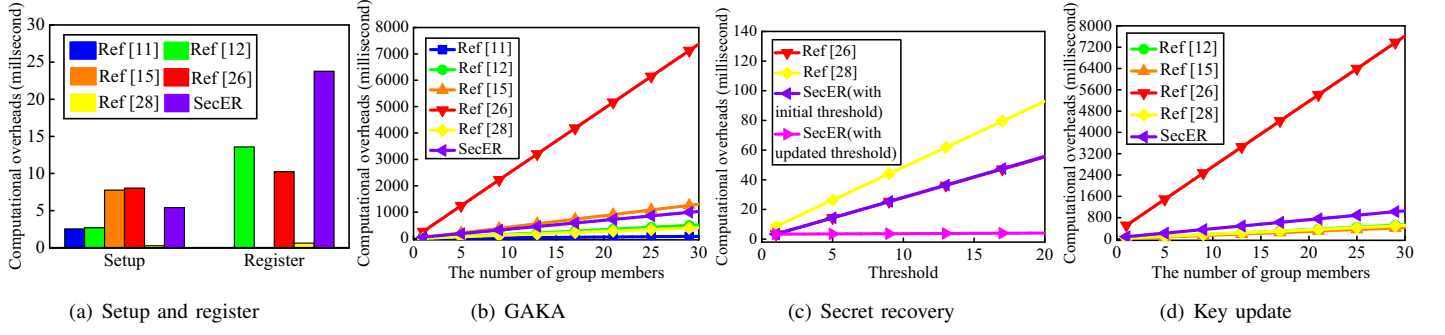


Fig. 6. Computational overhead comparison

Nevertheless, there are still gaps in real-life emergency rescue. At first, machine learning-based detection algorithms serve as a fundamental step in the secure communication of UAVs, so further work is to train the model that can identify unknown attacks, not just known ones. Secondly, blockchain has the potential to reform UAV-assisted IoVs, whereas consensus with high latency, low throughput, and non-scalability is unacceptable for time-sensitive emergency rescue. Therefore, further research is required on how to improve consensus efficiency under large-scale UAVs. At last, as a communication medium between UAVs and the ECV, the gateway may be captured causing UAV-assisted IoVs to be compromised even if terminals (i.e. UAVs and the ECV) are secure. Hence, it's inevitable to explore how to secure the

gateway between different hops.

VIII. CONCLUSION

For UAVs-assisted IoVs emergency rescue, we design an accountable GAKA protocol with changeable thresholds and verifiable shares supporting adaptive group members and updatable keys. Further to say, based on the pseudonym mechanism, random parameters, verifiable parameters, and updatable keys, our SecER realizes secure, resilient, reliable, and feasible multi-task (i.e. communication, computation, and sensing) emergency rescue. Additionally, our SecER satisfies semantic security and 12 design goals related to security (see the security proof and security analysis). Compared with state-of-the-art studies, our SecER's storage cost is slightly higher,

TABLE VII
COMPARISON OF COMPUTATIONAL OVERHEADS (IN MILLISECOND)

Scheme	Setup	Register	GAKA	Changeable thresholds	Verifiable shares	Secret recovery	Key Update
Ref [11]	T_{G_4-Mul}	T_h	$(9n + 8)T_h + (2n + 2)T_{prf} + (3n + 2)T_{La} + (n + 1)T_{G_4-Mul} + T_{Lookup}$	—	—	—	—
Ref [12]	T_{G_1-Mul}	$2T_{G_1-Add} + 5T_{G_1-Mul} + 2T_h$	$(4n - 3)T_h + (6n - 4)T_{G_1-Mul} + (n^2 + 3n - 2)T_{G_1-Add} + (n - 1)T_{Sig}$	—	—	—	$(4n + 1)T_h + (6n + 2)T_{G_1-Mul} + (n^2 + 5n + 2)T_{G_1-Add} + nT_{Sig}$
Ref [15]	$2T_{G_4-Mul} + T_{G_3-Mul} + 3T_h$	—	$3nT_{bp} + (2n - 1)T_{G_3-Mul} + (3n - 2)T_{G_2-Mul} + 6nT_{G_4-Mul} + nT_{G_4-Add} + (4n - 1)T_{G_3-Add} + (n + 1)(T_{mod} + T_{exp}) + (10n + 1)T_h$	—	—	—	$3nT_h + 5nT_{G_3-Mul} + 4nT_{G_3-Add} + (n + 1)T_{mod} + T_{G_4-Mul}$
Ref [26]	$3T_{G_3-Mul}$	$T_{G_3-Mul} + T_{mph}$	$(15n + 2)T_{bp} + (18n + 1)T_{mph} + 16nT_{G_3-Mul} + 5nT_{G_3-Add} + 2nT_{exp}$	—	—	$t(T_{G_1-Add} + T_{G_1-Mul}) + T_{mod}$	$(15n + 18)T_{bp} + (18n + 19)T_{mph} + 16(n + 1)T_{G_3-Mul} + 5(n + 1)T_{G_3-Add} + 2(n + 1)T_{exp}$
Ref [28]	$T_{che} + T_h$	T_{mod}	$2n(16T_{che} + 17T_h + 4T_{en/de})$	T_h	—	$2(2t + 3)T_{che} + (t + 3)T_{en/de} + (2t + 3)T_h + t(T_{G_1-Mul} + T_{G_1-Add}) + T_{mod}$	$2(n + 1)(16T_{che} + 17T_h + 4T_{en/de}) + (n + 1)(T_{G_1-Mul} + T_{G_1-Add}) + T_{mod}$
SecER	$2T_{G_1-Mul}$	$6T_{G_1-Mul} + 3T_h + T_{mph}$	$(3n + 1)T_{bp} + (2n + 1)T_{mph} + (n + 3)T_{G_1-Mul} + (n - 1)(T_{G_1-Add} + T_{G_2-Mul})$	$m(T_{mod} + mT_h)$	$n(T_{bp} + tT_{exp} + T_h + (t - 1)T_{Z_q^*-Mul})$	$t(T_{G_1-Add} + T_{G_1-Mul} + T_h) + T_{mod}$ or $mT_{G_1-Add} + T_{mod} + T_{G_1-Mul}$	$(3n + 4)T_{bp} + n(T_{G_1-Add} + T_{G_2-Mul}) + (2n + 3)T_{mph} + (n + 4)T_{G_1-Mul}$

n is the number of group members.

m is the updated threshold.

t is the initial threshold.

— is there is no such phase.

whereas the group members (i.e. UAVs) have very little communication overheads. Furthermore, the sum communication of group manager (i.e. the ECV) in the setup and register phases is reduced by 30% on average. The communication overhead of group manager in the GAKA and key update phases are respectively reduced by 1.744% and 12.44% on average. In short, our SecER has more preponderances in balancing security and availability.

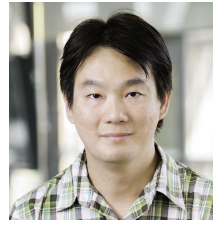
In future work, we chew over designing an AKA protocol supporting heterogeneous cryptosystems since network entities are run by various operators in real-world environments.

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